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ARTICLE

Global competition, local unions, and political representation: Disentangling mechanisms

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Abstract

While recent scholarship has demonstrated multiple political effects of international trade, less attention has been paid to unbundling the mechanisms through which import competition affects democratic politics. One mechanism, in theory, works through labor unions as domestic countervailing powers shaping legislative responses on compensation and trade votes. We assess the relevance of unions as a mediating variable in the US Congress. For identification, we leverage two distinct sources of exogenous variation, one instrument for import exposure and another for unionization, and combine them in a semiparametric estimator. We find that (i) import competition lowers district-level unionization, (ii) weaker unions lead to less legislative support for compensating economic losers and less opposition to trade deregulation, and (iii) the union mechanism represents a large fraction of the overall effect of import exposure on legislative votes. The results help explain weak compensation and further trade liberalization in the face of rising global competition.

The success of democratic capitalism rests on its ability to accommodate the sharp distributive conflicts resulting from exposure to global markets and technological change. When international trade lowers the wages or destroys the jobs of a significant number of workers, compensating trade losers through social insurance, retraining, and income redistribution can enhance efficiency, dampen economic inequality, and may even limit dissatisfaction with democracy. However, research in political science and economics demonstrates that compensation is not automatic. Winners from economic change, and the politicians representing them, may have few incentives to allocate resources to economic losers and possess the political power to block such policies.

To the degree that unequal economic resources entail unequal political influence biased toward economic elites and corporations, countervailing powers, which reduce this bias, help to make democracy work. Labor unions remain a major countervailing power for non-elite workers.[1] Not only do unions increase wages (Ahlquist, 2017; Farber et al., 2021), they also reduce unequal political responsiveness in a new gilded age characterized by high economic inequality in developed economies (Becher & Stegmueller, 2021; Flavin, 2018). But what if economic shocks that increase the need for policies to support economically disadvantaged citizens simultaneously undermine the labor unions that give them voice? Such a development would further weaken the political representation of trade losers and undermine the system of “embedded liberalism” forged after World War II, which combines trade openness with social protection (Mansfield & Rudra, 2021; Ruggie, 1982). Disillusionment with the promise of democracy and growing receptiveness to identity politics and populism may follow (Bisbee et al., 2020; Rodrik, 2018).

The dramatic rise of import competition from China and other developing countries since the 1990s has

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The Cornell Center for Social Sciences verified that the data and replication code submitted to the AJPS Dataverse replicates the numerical results reported in the main text of this article.

[1] For example, Galbraith (1952) conceives of labor unions, which are voluntary organizations that represent workers in collective bargaining with employers, as the canonical countervailing power in the labor market. We focus on the political marketplace.

magnified the challenges faced by democratic capitalism. It has also provided new opportunities for research on how global markets shape domestic politics. Building on research that studies its economic effects (e.g., Autor et al., 2016), a burgeoning literature has studied the political effects of the China shock (Autor et al., 2020; Ballard-Rosa et al., 2021; Bisbee et al., 2020; Bisbee & Peter Rosendorff, 2021; Colantone & Stanig, 2018a, 2018b; Feigenbaum & Hall, 2015; Kim & Pelc, 2021; Kuk et al., 2018). A seminal study analyzing the responses of elected policymakers to import competition from China revealed that legislators from more trade-exposed districts were more likely to take the protectionist side on trade votes (Feigenbaum & Hall, 2015). While not separately looking at policies that compensate trade losers, it finds that, on average, legislative votes on non-trade issues were not affected.

However, this literature has paid less attention to institutional mechanisms in general and labor unions in particular. A recent review concludes that the mediating factors linking import shocks to political outcomes "are still very much a mystery" (Frieden, 2022, p. 588). We address this gap and analyze the role of unions as a mediating variable that shapes legislative responses to global economic shocks and is itself changed by rising import exposure. We test the theory that labor unions are a major missing link between international markets and domestic politics in the American political economy. Following Feigenbaum and Hall (2015) and others, we focus on the US House of Representatives and study the impact of the shock on legislative votes on economic issues.

In line with political economy theories, we distinguish between trade and compensation votes. Our main theoretical and empirical departure from recent work on the legislative effects of trade (e.g., Feigenbaum & Hall, 2015; Owen, 2017) and the separate literature on the legislative impact of unions (e.g., Becher & Stegmueller, 2021; Becher et al., 2018) lies in our focus on disentangling and quantifying mechanisms through which import competition shapes political representation. From the extant literature, it is not obvious that the union mechanism will be substantively important. Trade may not have more than a minor effect on union strength (Ahlquist & Downey, 2023; Scruggs & Lange, 2002), and even if it does, unions might not be able to shape legislative votes on key economic issues. Even if they do, their mediating impact may be dwarfed by other mechanisms. For example, a canonical view of democratic politics, formalized in political agency models, suggests that reelection-seeking policymakers may seamlessly respond to constituencies hit by economic shocks (for evidence, see Mian et al., 2010; Owen, 2017). Moreover, other mediating institutions, such as the media (Snyder & Stromber, 2010), may be more important than unions. We provide clear

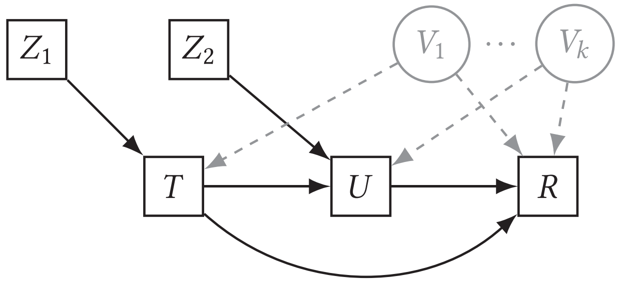

FIGURE 1 Decomposing the effect of import shocks on congressional representation. *Notes*: How important is the union mechanism? This figure illustrates the causal mediation identification problem and how it can be addressed using two instrumental variables. The goal is to decompose the impact of import competition (T), the treatment, on legislative votes on compensation and trade (R) into a direct effect and an indirect effect via labor unions (U), the mediator. If there are unobserved confounders ($V_1, \ldots, V_k$), the decomposition needs two instruments: one to shift trade exposure (T), another to shift unionization (U).

empirical evidence for the importance of unions as a mechanism.

Of course, analyzing causal mechanisms is difficult (Imai et al., 2010, 2011). It requires analysts to deal with a double threat of confounding. Figure 1 illustrates the problem and our research design. First, estimates of the total effect of the treatment, import exposure (T), on political representation in terms of legislative (or roll-call) votes (R) will be biased if there is an unmeasured confounder (V_1) that shifts both import exposure and representation. For example, politically influential (e.g., swing) districts may be better protected from import competition because they have more political weight and, for the same reason, are better represented on numerous other issues. Absent experimental manipulation, these endogeneity problems can be addressed using an instrumental variable (IV; i.e., Z_1). The literature has identified plausibly exogenous instruments for import exposure based on import exposure in the same industries in other developed economies (Autor et al., 2013a, 2016).

Importantly, having an instrument for import exposure alone is not sufficient to decompose the effect of import exposure into an indirect effect, working through unions, and a direct effect. The second confounding problem stems from the fact that one cannot independently randomize union strength, our mediating variable. It is not difficult to think of historical factors and institutions that may shape both unions and policymaking (Ahlquist, 2017). For instance, historically, race has been used as a tool to divide workers, prevent the growth of unions, and dampen support for compensation. State-level laws regulating unions and collective bargaining tend to be more restrictive in the US South (Feigenbaum et al, 2018; Flavin & Hartney, 2015), and some scholars argue that legacies of slavery still explain differences in preferences and political behavior within states (Acharya, Blackwell & Sen, 2019). Moreover, correlated voter preferences may explain lower demand for unions in the workplace and less support for policies favored by unions. In general terms, Figure 1 captures that there may be confounding variables (V_k) that shape union strength and congressional representation. This makes it challenging to identify the effect going from U to R. Therefore, we also need an independent IV for district-level union membership (Z_2). Turning to a separate literature, we draw on history and leverage across-the-board unionization in coal and metal mines and steel plants just after World War II as a source of plausibly exogenous variation (Becher & Stegmueller, 2021; Holmes, 2006). To combine both instruments without making strong functional form assumptions, we employ a novel semiparametric IV model (Frölich & Huber, 2017). In sum, our approach leverages two natural experiments and advances in statistical methods for causal mediation analysis. While we cannot conduct a parallel experiment randomizing both trade exposure and union membership at the scale of congressional districts, our approach enables us to relax strong selection on observables and functional form assumptions required in the standard regression approach for unpacking causal mechanisms.

Altogether, our analysis makes two primary contributions. First, we advance the large literature on how institutions in general and labor market institutions in particular condition the impact of international economic shocks on domestic politics (Garrett, 1998; Guiso et al., 2019; Milner & Kubota, 2005; Rudra, 2008). Our empirical approach captures that conditioning institutions may themselves be altered by trade shocks and operate as a mediating force, enabling us to unbundle direct and indirect effects on political representation. Combining arguably exogenous variation for both international shocks and trade and unions, we are the first to quantify how much of the impact of rising import competition on legislative responses on compensation and trade policies works *via* the adverse effects of import competition on unionization. We estimate that about one half of the effect of import exposure on declining legislative support for compensation works through the channel of weaker unions. On trade policy, unions account for approximately one third of the total effect in the direction of less opposition to further trade liberalization. Our results help explain the otherwise surprising lack of support by legislators from highly trade-exposed districts on legislative votes about compensation. They also highlight that import competition can increase legislative support for hyperglobalization by weakening unions.

Second, our analysis speaks to long-standing research on the consequences of globalization for organized labor and the recent literature on the economic effects of the China shock. Recent scholarship has shown how import competition affects labor market outcomes like wages and employment, though it has paid less attention to unions (for a review, see Autor et al., 2016). The first step of our analysis reveals that import competition has pronounced effects on local labor unions. While the hypothesis that global markets undermine domestic commitments to compensation by weakening unions is not new, the earlier (i.e., pre-China shock) literature on the link between globalization and unions focused on a period when most trade was between developed economies and it largely relied on country-year–level analyses (for an exception, see Slaughter, 2007). It suggested that globalization was no smoking gun for explaining union decline (Garrett, 1998, p. 62; Scruggs & Lange, 2002). However, patterns of world trade have since changed. One of the few studies on unions in the China shock literature finds that import exposure is linked to a modest decline in unionization in manufacturing industries but not to overall de-unionization at the state level (Ahlquist & Downey, 2023).[2] We leverage *within-state* variation in import exposure using administrative data and find that import exposure since the 1990s has diminished union membership at the district level.[3]

GLOBAL COMPETITION AND ENDOGENOUS COUNTERVAILING INSTITUTIONS

We consider labor unions as an endogenous countervailing power in the democratic politics of compensation and trade policy. They enhance the political representation of economic losers from global competition. But they are not immune to the economic effects of globalization. Therefore, it is key to explore the relevance of unions as a mediator between import competition and the political representation of economic losers.

Labor unions and political representation

Labor unions matter for the political representation of non-elite workers, whether they are unionized or not. First, unions shape policy preferences. They push back against the fatalistic view that the negative distributive effects of globalization are inevitable or even

[2] Charles et al. (2021) find a negative effect on certification elections.

[3] There are multiple reasons why our results may be different from Ahlquist and Downey (2023), including the focus on different geographic units (districts vs. states) and the use of administrative data to measure union membership, which entail different strengths and weaknesses compared to survey data.

necessary collateral damage on the road to greater aggregate prosperity. Through leadership and socialization, unions tend to promote the view that governments can and should take actions to help economic losers and reduce economic inequality. Accordingly, economic losers deserve collective solidarity as people's economic fate is in part determined by circumstances and luck, not just hard work and competence (Ahlquist & Levy, 2013; Mosimann & Pontusson, 2017).

American unions regularly take policy positions in support of the less affluent and those more exposed to labor market risks, through policies that compensate for or insure against income loss and maintain the required revenues through taxes. They are also more critical of free trade. In what is probably the most comprehensive study of the nexus between public opinion and interest group positions covering hundreds of issues, Gilens (2012, p. 158) finds that unions in the United States exhibit a "strong tendency to share the preferences of the less well-off," more so than any other group analyzed. These positions are broadly consistent with self-interest, other-regarding considerations, and ideology. Unions in trade-exposed local labor markets have an interest in providing a collective voice in support of policies that benefit their members' economic interests. For instance, this includes support for unemployment insurance, worker training, and income redistribution more generally. Recent micro-level evidence is consistent with the argument that norms of solidarity, often invoked by leaders, shape the relatively high support among union members—even those in the upper half of the income distribution or working in competitive exporting firms—for income redistribution and protectionist measures (Ahlquist et al., 2014; Kim & Margalit, 2017; Mosimann & Pontusson, 2017).

Second, through electoral mobilization unions shape the outcome of elections, potentially tilting the balance away from economic elites and big business. Over decades, a cross-disciplinary literature has documented union effects on elections and policymaking (Ahlquist, 2017; Freeman & Medoff, 1984; Leighley & Nagler, 2007). Relatedly, in the marbled halls of power, unions are one of the few lobbying groups speaking on behalf of non-elite workers, and stronger district-level unions are associated with lower bias in the perceptions of congressional staffers of their district's support for redistributive policies (Hertel-Fernandez et al., 2019). Overall, organized labor has anchored the Democratic Party and made it more attentive to economic losers (Schlozman, 2015). While earlier studies often lack credible causal identification (Ahlquist, 2017), recent research is increasingly based on natural experiments and other designs that make it more plausible to infer union effects on democratic politics.

Import competition and unionization

The crux is that labor unions are themselves directly subjected to the forces of globalization. On the one hand, increasing exposure to economic risks in the global economy may increase demand for unions. On the other hand, international trade and, in particular, import exposure can weaken unions by undermining their position at the bargaining table, eliminating entire unionized workplaces, and debilitating efforts to organize non-union workplaces in other sectors. While cross-national research has yielded mixed findings about the impact of economic globalization on unionization and soundly rejected the idea of an inevitable race to the bottom with respect to labor market institutions, the institutional environment makes American unions vulnerable to import competition.

Even in the absence of large-scale job losses, import competition can contribute to declining unionization by weakening unions' bargaining power and eroding the benefits of union membership. In standard models of wage bargaining between profit-maximizing firms and unions, an increase in international product market competition lowers firms' quasi-rents up for negotiation, leading to deteriorating outcomes for unions at the bargaining table (Rodrik, 1997, chapter 2). Crucially, lower bargaining power and declining union wage premiums make it less attractive for workers to join a union and more difficult for unions to recruit new members.

Moreover, recent evidence suggests that import exposure has reduced manufacturing employment and, through general equilibrium effects, negatively affected local labor markets (Autor et al., 2013a). The resulting job losses, plant closures, and declining outside options exacerbate the organizational problems faced by unions. The comparatively combative nature of wage-setting at the establishment level (rather than at the industry level) gives firms incentives to take a more hostile stand against unions and leverage dismissals and the threat of plant relocation to weaken organized labor (Scruggs & Lange, 2002).

Trade-induced plant closures exert additional downward pressure on unions. Plant closures account for a significant share of the decline in manufacturing employment (Fort et al., 2018). This matters because local unions are formed at the establishment (i.e., plant) level. Federal legislation implies that any new establishment is union-free by default, even if other employees in the same company are unionized. Within the institutional framework supervised by the National Labor Relations Board, forming a union for collective bargaining with the employer is costly as it requires first, collecting a sufficient number of signatures to request a certification election for the new bargaining unit, and, if this is achieved, winning a majority in the election against the (often substantial) opposition of management. Reinforcing this logic, companies that shut down an existing plant may pursue a strategy of domestic offshoring and open a new plant in a different state where it is more difficult to unionize. Altogether, the dynamics of wage bargaining under increased international competition as well as trade-induced job losses and plant closures imply that import competition leads to a decline in the stock of union members.

Social spillovers may be a mitigating force. Holmes (2006, p. 1) argues that "unionism is contagious at the geographic level" (cf. Ahlquist & Downey, 2023). Union members tend to have more positive views of organized labor than comparable nonmembers, due to socialization or experiencing the benefits of membership. When changing jobs, union members will take their taste for unions with them. Furthermore, union preferences can be transmitted to children working in other sectors. In line with this argument, Holmes (2006), using contract-level data, finds that proximity to mining and metal industries in the middle of the twentieth century is strongly correlated with unionization in health care facilities five decades later. The study also finds evidence of positive spillovers from mining and steel to retail and construction. However, theoretically it is not clear that spillovers can fully compensate for the decline in unionization stemming from trade-induced declines in bargaining power, job losses, and plant closures. But, as explained in more detail below, we argue that geographic spillovers before the dramatic change in world trade in the 1990s constitute a key source of identification for our effect decomposition.

Empirical implications

We empirically assess the following mechanisms linking import competition at the level of congressional districts to legislative votes on compensation and trade policy of the member representing each district. Districts are the theoretically appropriate unit of analysis for the purpose of studying the causal chains from international trade to legislative votes. They are also attractive from an inferential perspective because much of the variation in import exposure from China stems from different local labor markets in the same state (Autor et al., 2013a) and a within-state analysis can better untangle import competition from exposure to technological shocks (Autor et al., 2013b).

- First, higher import exposure reduces unionization.
- Second, stronger unions increase legislative support for compensation and reduce legislative support for further trade liberalization.
- Third, we ask how important the union mechanism is relative to the total effect of import exposure on legislative votes. If unions are an important countervailing power that is undermined by global competition, the channel operating through district-level unions should be substantively important.

DATA AND INSTRUMENTAL VARIABLES

We examine the impact of imports from China between 1990 and 2000 on subsequent political representation via union membership and other channels in electoral districts for the House of Representatives during the 107–112th Congress (i.e., 2001–2012). During the 1990s, imports from China rose by 95.3 billion (in 2007 US$) or 362% (Autor et al., 2013a, p. 2131). Subsequently, Congress considered numerous policies concerning compensation and trade.

Roll-call votes on trade and compensation

We examine high-profile votes on trade policy and compensation policies, which are conceived to include income protection and social insurance in the face of labor market risks (Garrett, 1998; Rodrik, 1997). On trade, we have 10 votes. Following the selection of Owen (2017 p. 302), this includes votes on free-trade agreements and the vote on granting the president fast-track authority to negotiate trade agreements (which was opposed by unions). We added three more recent free-trade votes not covered by Owen's study. Based on interest group ratings from the Cato Institute, voting yes on these votes means supporting trade liberalization.

On compensation, we select 13 key votes. This includes votes to extend unemployment benefits for people who have exhausted regular benefits, and expand job training, housing assistance, food stamps, or increase the minimum wage, and votes on broader increases in social expenditures or tax cuts financed by cutting future social spending, as identified by the legislative scorecard of Americans for Democratic Action.

These are policies that mitigate economic hardship in areas affected by international competition (Autor et al., 2013a, p. 2149). We also include three House votes on the extension of trade adjustment assistance that took place during the period of study. All compensation votes are coded so that 1 means supporting the pro-compensation direction and 0 means opposing it. For details on included votes, see Supporting Information (SI) Section A (pp. 2–3). While these votes do not exhaust the issues that are salient to unions, they match our theoretical focus on the politics of trade and compensation.[4]

[4] Our analysis does not take a stand on whether protection and compensation are substitutes (cf. Kim & Pelc, 2021). In theory, compensation for increased openness is more efficient than protectionism, but promises of sustained

Import exposure

To measure a district's exposure to import competition from China, we follow the well-known approach pioneered by Autor et al. (2013a, 2016). The idea is to map industry-specific import shocks onto local labor markets in proportion to the localities' pre-shock industrial employment structure. The shift-share measure captures the change in Chinese import exposure per worker in a local labor market apportioned by its share in total national employment. The measure from Autor et al. (2013a) covers 741 local labor markets (commuting zones, each consisting of a group of counties). We capture the exposure of a district's population to import shocks, by creating a mapping from commuting zones (in their 1990s definition) to congressional districts, based on the spatial overlap of the commuting zone and district weighted by the population in the spatial intersection for *each* Congress (see SI B, p. 4, for more details).[5]

Instrumenting import exposure

Imports from China are possibly related to unobserved district-level characteristics that also drive unionization or legislative votes. To address these endogeneity issues, we follow Autor et al. (2013a) who construct an instrument from Chinese imports to other high-income markets. Their patterns of import growth across industries are highly correlated with those in US industries (Autor et al., 2016, p. 219) and form a strong instrument plausibly independent of local factors in congressional districts.

Previous work shows that this instrument has a strong first-stage impact on realized import exposure in American commuting zones (Autor et al., 2013a). We verify that the same holds true at the level of congressional districts: A 1% increase in the instrument induces a .83% (± .05%) increase in import exposure (SI B, p. 4). Recent methodological work on shift-share designs clarifies the identification challenges for a region-level analysis of trade shocks, focusing on the plausibility of assuming exogeneity of either the shocks (Borusyak et al., 2021) or the shares (Goldsmith-Pinkham et al., 2020). The latter requires that pre-shock employment shares are conditionally exogenous (while the trade shift may be endogenous). Their analyses (albeit for a different outcome) suggest that this may be implausible, and they provide strategies to assess the sensitivity of IV results to overly influential individual industries. While we discuss this issue in more detail in SI Section C (p. 10), we outline here four ways in which we tackle this issue. First, as suggested by both studies, we control for technological change. Second, as discussed later, we adopt a flexible approach to make conditional exogeneity more plausible. Third, following Goldsmith-Pinkham et al. (2020), we calculate Rotemberg weights and exclude industries with the largest weights from the analysis (SI Table C.1 and SI Table G.1). Finally, we show that our trade effect estimates can also be replicated with other measures of exposure (SI Table C.3).

Union membership

To measure union membership (as a share of the working population), we rely on fine-grained administrative union data collected by Becher et al. (2018). They use Labor Management (LM) forms—mandatory annual reports filed by each local union to the Department of Labor—providing more than 350,000 individual reports covering almost 30,000 unions. The data include membership size and geocoding information for each union, which were used to map unions onto congressional districts.

Instrumenting union membership

As already discussed, a key challenge for our analysis is that unionization is not necessarily exogenous to factors (e.g., preferences, policies, institutions) that shape political representation. We thus employ an IV for district-level unionization. The history of organized labor combined with geography suggests a plausibly exogenous source of variation in contemporary union membership that we can use as an instrument in our analyses. Following Holmes (2006) and Becher and Stegmueller (2021), we leverage record-high unionization levels in coal and metal mines as well as steel plants during the heyday of unionism just after World War II as a plausibly exogenous source of variation. The instrument rests on the argument that initial unionization in mining and steel spilled over into stronger unions in other sectors, including nonmanufacturing sectors like health care and groceries, through socialization and union organizational efforts. Holmes (2006) provides detailed evidence on spillovers from mining and steel in the 1950s into nonmanufacturing unionization decades later. Building on this finding, Becher and Stegmueller (2021) use the historical employment share in mining and steel as an instrument for contemporary union

compensation in exchange for openness may not be credible. Our analysis takes place in a time of low trade barriers, where opposition to further integration (e.g., with Chile) need not benefit current losers from Chinese import competition.

[5] Because commuting zone geographies serve as a time-constant reference point, this approach captures decennial redistricting in each apportionment cycle and mid-period court-ordered redistricting. The unit of analysis in the decomposition is the district-vote, where districts are stacked over congressional terms.

FIGURE 2 Instrumenting union density with post-war mining/steel industry employment. *Notes*: Panel (a) plots logged district-level employment in mining and steel industries (as a share of working population) in 1950 against logged contemporary district-level union density (share of union members among working population) in the 109th Congress. Panels (b) and (c) plot district-level employment in mining and steel industries in 1950 against US district-level import exposure (b) and its instrumental variable (IV) in panel (c). Linear fit with cluster-robust 95% confidence interval superimposed.

strength. Importantly, the location of mining and even steel production is mainly determined by nature, based on availability of coal and ore in the ground or the access to raw materials for steel mills. In the 1940s and 1950s, coal, metal, and steel were unionized across the country, regardless of states' political leaning and prior strength of the labor movement (Holmes, 2006, pp. 3–5). The combination of natural origins and spillovers implies that part of the variation in union membership in congressional districts at the beginning of the twenty-first century is exogenous to local factors conditional on the intensity of mining and steel employment in the middle of the twentieth century.[6]

We show this relationship graphically in panel (a) of Figure 2. It plots logged district-level employment in mining and steel industries (as a share of working population) from the 1950 Census against logged contemporary district-level union density (share of union members among working population) in the 109th Congress. The plot reveals a clearly positive relationship that is statistically and substantively significant. A simple linear model (with cluster-robust s.e.) reveals that a 1% increase in 1950s employment in mining and steel induces a .32% (± .05%) increase in the district-level share of union members. Thus, we use district-level variation in mining and steel employment in the 1950 Census as an instrument for present-day unionization levels.

As our main analysis combines both IVs to disentangle the relative importance of the union mechanism from other mechanisms through which import competition shapes congressional representation, we have to verify that the two instruments are separable (conditional on covariates). If the two instruments were closely related, they would not serve our effect decomposition. One possibility is that historical mining and steel employment drives exposure to import competition, perhaps because stronger unions got more trade protection (Ahlquist & Downey, 2023; Owen, 2013). While this would not invalidate identification as long as the instrument for unionization remains uncorrelated with the instrument for import exposure, it raises an important concern about temporal sequencing that is assessed empirically in panels (b)

[6] Given the theory behind the instrument, we prefer an instrument that captures the number of union members. Spillover effects are more pronounced when more individuals are members of a union in an area. This is why we use employment shares in mining and steel plants as opposed to, say, data on mineral deposits or a simple count of mines. A weakness of this approach might be that employment shares are partly determined by endogenous factors (e.g., larger mining operations might occur in regions that have a larger industrial base or more entrepreneurial environment). While we cannot fully rule out these issues, we provide a robustness test in which we account for existing historical (1950s) shares of industrial employment in a region. Furthermore, our empirical analyses also include hypotheses tests that assume a degree of exclusion restriction violation.

and (c) of Figure 2.[7] We see that historical mining and steel employment at the district level in 1950 is not associated with actual import exposure in the 1990s (panel b). This pattern is consistent with the argument and evidence in Holmes (2006) that unionization in mining and steel during the historical peak of unionization spilled over into other sectors of the economy, including services (e.g., health care and groceries), that gain from cheaper imports. Importantly, there is only a weak (nonsignificant) association between the two instruments (panel c): An increase in mining and steel employment in 1950 is associated with a −.03 (± .05) percentage point decrease in the import exposure instrument.

EMPIRICAL STRATEGY

Instrumental variable models

Before turning to the effect decomposition, we separately assess (i) the effect of import exposure on unions and (ii) the effect of unions on legislative votes using IV analyses. Using the instruments introduced above, our preferred specification employs a semiparametric IV model that relaxes functional form assumptions for control variables.

Depending on the analysis, the outcome will be either district-level union membership or votes on compensation and trade by the district's representative in each Congress. In each analysis, the causal or treatment variable of interest, import exposure, or unionization, is instrumented with the instruments discussed above. While control variables are not needed when the instruments are valid, exogeneity of the instruments may be more plausible conditioning on pre-exposure controls. The appeal of using debiased machine learning for the IV models is that it is more robust against misspecification of the functional form of the controls, as it can flexibly account for higher order interactions between control variables and for nonlinear terms.

The set of possible district-level control variables includes district characteristics measured from the 1990 Census (share of college degrees, share of Black and foreign-born population, urbanization, manufacturing employment), before the trade shock. To capture possible legacies of slavery and immigration (Acharya et al., 2019), we also include variables from the 1910 Census (share Black and foreign-born), deliberately measured before the great migration. We also account for pre-China shock susceptibility to automation because sector-biased technological change may be correlated with import exposure and the strength of unions. Instrument validity would be threatened if changes in automation drive import exposure the same way across countries. While in smaller geographic units, such as commuting zones or districts, the susceptibility to automation and import competition are only moderately correlated (Autor et al., 2013b; Autor et al., 2016), we control for occupational routine task intensity (Autor et al., 2013b) matched to districts, resulting in a measure of the fraction of the workforce in the district at the risk of automation. To proxy for political factors not accounted for by state-fixed effects and other district characteristics, our specifications also include districts' political ideology in the 1980s based on Kernell (2009). SI Table B.1 provides descriptive statistics. When included, controls are normalized to have mean zero. Additional controls are included in robustness checks (reported in SI C, pp. 7–9, and SI D, pp. 12–13).

Following the strategy outlined in Chernozhukov et al. (2018), the set of controls can flexibly enter both the first stage, where the treatment variable of interest is regressed on the instrument and controls, and the second stage, where the outcome variable is regressed on the instrumented treatment variable and controls, via possibly highly nonlinear functions. We estimate these using debiased machine learning for IV (dML-IV). See SI Section F (pp. 15–17) for technical details.

Effect decomposition using two instruments

Our central quantities of interest are the indirect effect of trade shocks on legislative votes, via unions, and its magnitude relative to the direct effect. Figure 1 summarized the theoretical argument that trade (T) impacts union membership (U), which, in turn, affects legislative (i.e., roll-call) votes (R) and the identification problem. Using standard terminology, the full causal path working through unionization, $T \rightarrow U \rightarrow R$, is called the *natural indirect effect* (NIE). Trade may also have a *natural direct effect* (NDE) on roll-call votes, representing channels other than union density linking the shock to votes. For example, politicians may respond to voter demand regardless of union strength.

Recent work has explicated the challenge for such an effect decomposition. Intuitively, it entails a double confounding problem (called sequential ignorability by Imai et al., 2010; Imai et al., 2011). Following the identification strategy of Frölich and Huber (2017), we combine our two IVs, one for trade exposure and another for unionization, in a single statistical model that enables us to unbundle the total effect of import exposure under weaker assumptions compared to the classical approach requiring selection on observables and linear functional form assumptions. In the spirit of

[7] Furthermore, we provide robustness tests that adjust for pre-shock industry-level unionization estimated from survey data.

IV designs for nonexperimental data, we think of our approach as replacing implausible assumptions about selection on observables and known functional forms with more plausible (but not infallible) assumptions leveraging natural experiments and advances in the statistical analysis of effect decomposition.

To clarify the counterfactual quantities we want to estimate, let us denote two counterfactual levels of trade exposure T as t and t' (omitting subscripts). Then the potential vote outcome for an observation under some hypothetical trade exposure treatment is represented by $R(t,\ U(t'))$. This signifies that a roll-call vote is driven by (non)exposure to a trade shock as well as union density, $U(t')$, which is itself a function of the shock. The NIE of the trade shock is obtained by holding the exposure to the shock constant in both counterfactual scenarios but shifting union density from its level with without trade exposure $U(0)$ to its level when exposed to the shock $U(1)$. The difference between the two terms is the impact of the trade shock on roll-call votes due to the change in union density:

$$E\left[R(1,\ U(1)) - R(1,\ U(0))\right] \tag{1}$$

The NDE of the trade shock is obtained by holding union density constant while shifting the potential outcome from the value it takes absent trade exposure to the value it takes when exposed to the trade shock:

$$E\left[R(1,\ U(1)) - R(0,\ U(1))\right] \tag{2}$$

The direct effect thus indicates the total effect of the trade shock on roll-call votes net of the effect the trade shock has on votes via changes in union density. For the analysis, we define an "intensive trade shock treatment" as occurring whenever a district's import exposure exceeds the median of the district distribution, which corresponds to about \$1,000 per worker.[8]

The identification problem of such a causal structure arises from the likely existence of unobserved confounders ($V_1,\ \ldots, V_k$). Even in the ideal scenario of controlled random assignment of T (or a credible instrument for it), a classical decomposition does not provide a valid estimate of the indirect effect—one can still only identify the causal effects of T on U and T on R, but not the causal effect of T on R *via* U. Our empirical strategy to address the problem employs *two* instruments to identify the indirect effect of trade exposure, as illustrated in Figure 1. It follows the approach proposed by Frölich and Huber (2017). The role of the first instrument, Z_1, is to generate an exogenous shift in the treatment. In our case, this is the instrument shifting US imports from China based on Chinese imports in other high-income markets. The second instrument, Z_2, enables us to vary union density independently of trade shocks. In our application, Z_2 is a district's mining and steel employment share in 1950. Going beyond the simplification in Figure 1, we do not need to assume constant effects (or the absence of treatment–mediator interactions). Our approach captures that there may be heterogeneity in the different causal chains, not only when estimating the total effect, as in standard IV models estimating local average treatment effects (LATEs), but also in the chain from import exposure to unions and unions to legislative representation.

As outlined in the section "Data and instrumental variables," Z_2 uses geographical spillovers from unionized mining and steel establishments in the middle of the twentieth century to other sectors of the economy in the same locality based on ongoing socialization and union mobilization efforts. The onset of the exogenous trade shock captured by Z_1 is more recent. While not instantaneous, trade effects driven by lower wages and job displacement will be visible within a few years. Using the first instrument, we can estimate the overall effect of import competition on congressional representation. The second instrument enables us to model the counterfactual comparisons represented by Equations (1) and (2). Under assumptions summarized below, we can thus decompose the total effect into an indirect effect, working through unions, and a remaining direct channel. Intuitively, the role of the second instrument is to undo the changes in U induced by T in order to identify the indirect effect $T \rightarrow U \rightarrow R$. For a detailed exposition of the model, assumptions, and estimation, see SI Section G.1 (pp. 18–20).[9]

A simplified example may convey how the two instruments help us to approximate the counterfactual NIE. Imagine that there are two districts, A and B, that are exogenously exposed to import competition, which weakens unions in both places. But the second instrument, based on plausibly exogenous spillovers from historical mining and steel realized before the trade shock in district B, shifts post-shock union density in B to the same level as the pre-shock union density in district A. In this comparison, the second instrument undoes the effect of import exposure on unionization, and we can compare legislative votes under high import exposure and varying union strength. Importantly, the post-treatment difference in unionization between A and B due to the second instrument is not a result of (observed or unobserved) confounders.

[8] That is, we set $T = 1$ ($IPW_d > Q_{\{IPW\}}(.505)$). As in IV approaches generally, effects are defined for the subpopulation of compliers, that is, districts whose import exposure is shifted upwards when import exposure in the same industries in other advanced economies moved above the threshold defined above (see SI Section G.1, pp. 18–21, for a more detailed discussion). Note that a more stringent threshold yields similar results (SI Table G.1).

[9] A placebo Monte Carlo study (SI G.5, pp. 23–25) indicates that the model adequately recovers null effects for the direct and indirect effect under substantial confounding.

TABLE 1 Effect of trade shocks on (logged) union membership.

	Instrumental variable (IV) (1)	IV (2)	IV (3)	IV (4)	Debiased machine learning IV (dML-IV)[a] (5)
Chinese import exposure	−.309	−.265	−.349	−.363	−.175
	(.045)	(.049)	(.036)	(.035)	(.035)
IV violation robust inference [p][b]	.000	.000	.000	.000	–
District characteristics:					
Slavery and immigration history		✓	✓	✓	✓
Demographics and ideology			✓	✓	✓
Technological change				✓	✓
Observations	2,592	2,592	2,590	2,554	2,554

Notes: Two-stage least squares (2SLS) estimates of IV models with state-congress fixed effects and cluster-robust standard errors (in parentheses). Chinese imports per worker is instrumented by imports to eight other highly industrialized countries. First-stage robust F statistics range from 307.7 to 465.6. Historical patterns of slavery and immigration proxied by 1910 Census district-shares of Blacks and foreign-born. A district's demographics include 1990s values of the share of foreign-born, Blacks, college-educated, employed in manufacturing, living in urban areas. District ideology is based on Kernell (2009) for the 1980s. Technological change is measured by risk of automation (RSH_d).

[a]Partial linear IV model estimated using debiased machine learning with sample splitting (100 sets) and fivefold cross-fitting (Chernozhukov et al., 2018). Estimates and standard errors are calculated using the median method. See Supporting Information (SI) Section F for details.

[b]Test of trade shock allowing for violation of IV exclusion restriction, H_0: $\beta = 0$ w. $Cov(Z,e) \approx 0$ (Wang et al., 2018). See SI Section E.

For the decomposition to have a causal interpretation, standard IV assumptions need to hold. The instruments must be relevant, plausibly exogenous, and satisfy an exclusion restriction. Drawing on the literature, evidence of a strong first stage, and additional robustness checks we argue that they are plausible. In addition, but closely related, the instrument for the trade exposure treatment (T) must be (conditionally) independent of the instrument for the mediator (U). As was already shown in Figure 2, the correlation between the two instruments is indeed small. Districts with higher mining and steel employment in the 1950s do not face substantively more (or less) potential import competition throughout the 1990s based on exposure in the same industries in other advanced economies. Finally, unobserved confounders (V_k), which can be thought of a latent variable, that shift the moderator should do so monotonically. The model is identified nonparametrically. It allows for arbitrary treatment–mediator interactions and thus heterogeneity in the direct and indirect effects of the trade shock on representation. While not certain, these assumptions are weaker than a standard linear structural equation approach requiring selection on observables and linearity.

RESULTS

Does import exposure reduce district-level unionization?

Table 1 shows IV estimates of the effect of Chinese import exposure during the 1990s on union density in House districts in the 107–112th Congress.[10] The most conservative estimate comes from the most flexible specification using the dML-IV on all district-level controls (column 5). It implies that a $1,000 increase in per-worker import exposure reduces unionization by .175 log points or about 16%. The other columns of Table 1 show the estimates from standard IV models, which adjust for state and time fixed effects to focus on within-state variation. State fixed effects account for time-invariant differences in institutions, including right-to-work legislation, and preferences across states. District controls enter sequentially. The resulting estimates are somewhat larger than those in column (5). One advantage of these models is that they allow us to conduct inference under some (local-to-zero) violations of the instrument exclusion restriction. As even small violations can lead to unreliable hypothesis tests, we employ a test based on the approach of Wang et al. (2018). The entries in Table 1 show p-values from an Anderson–Rubin test of the null hypothesis of no effect of trade on union density allowing for a local violation of the exclusion restriction (cf. SI Section E). Results indicate that small violations of the exogeneity of the trade instrument would not change our conclusions.

In all models, import exposure from China has a precise negative effect on district-level union membership. Additional robustness tests are reported in the SI. They include alternative specifications of district socio-demographics, adjusting for offshorability, for

[10] Alaska and Hawaii are excluded due to missing data. For comparability with later sections, Table I presents results from a stacked analysis that pools all congressional terms. All results go through in a congress-by-congress analysis (SI C.3, p. 10).

TABLE 2 Effect of union density on trade roll-call votes.

	Instrumental variable (IV) (1)	IV (2)	IV (3)	IV (4)	Debiased machine learning IV (dML-IV)[a] (5)
Union density [log]	−0.414	−0.382	−0.427	−0.463	−1.170
	(0.092)	(0.092)	(0.125)	(0.124)	(0.284)
IV violation robust [p][b]	0.000	0.000	0.006	0.003	–
Observations	4,243	4,243	4,243	4,183	4,183
District-level controls:					
Slavery and immigration history		✓	✓	✓	✓
Demographics and ideology			✓	✓	✓
Technological change				✓	✓

Notes: Two-stage least squares (2SLS) estimates of IV models with state-congress fixed effects and cluster-robust standard errors (in parentheses). Controls are as in Table 1. Union density is instrumented by share of mining employment in the 1950s. First-stage robust F statistics range from 14.0 to 33.7. Supporting Information (SI) Table I.1 presents full table including inference robust to potentially weak instruments.
[a]Partial linear IV model estimated using debiased machine learning with sample splitting (100 sets) and fivefold cross-fitting (Chernozhukov et al., 2018). Estimates and standard errors calculated using the median method. See SI Section F for details.
[b]Test of union coefficient allowing for local-to-zero violation of IV exclusion restriction, H_0: $\beta = 0$ w. $Cov(Z,e) \approx 0$ (Wang et al., 2018).

industry-specific union shares in 1990, instrumenting automation exposure, using post-2000 trade exposure, and a sensitivity analysis based on Rotemberg weights (SI Section C.2, pp. 7–9). To ensure that our results are not dependent on the trade instrument, SI Table C.3 reports results using (i) variation in tariffs across industries induced by China's entry into the WTO (Pierce & Schott, 2016) and (ii) residuals from a gravity model (Autor et al., 2013a). Invariably, these models reveal a negative impact of trade exposure on union density.

Do unions affect congressional votes on compensation and trade?

The second step in our analysis is to reassess whether unions have a causal effect on legislative votes on compensation and trade issues. The dependent variable in this analysis are roll-call votes of representatives. For trade issues, votes are coded so that 1 is a vote for trade liberalization and 0 is a vote against. For compensation issues, votes are coded such that 1 is voting in favor of expanding/protecting compensation and 0 is against. The key variable on the right-hand side of the regression is the (logged) share of union members among a district's working population instrumented using the share of employment in mining and steel in 1950.

The resulting estimates for trade roll-call votes are presented in Table 2 and the estimates for compensation roll-call votes are presented in Table 3. Across specifications, the estimates are consistent with the view that stronger unions lead to less legislative support for trade deregulation (Table 2) and more legislative support for economic compensation (Table 3).

In each table, column (5) reports results from the IV model estimated using debiased machine learning (dML-IV). The estimates imply that an increase in union density leads to a significant decrease in the likelihood of agreeing to trade roll-calls (-1.17 ± 0.28) and to a significant increase ($.38 \pm .18$) in voting for compensatory policies. In substantive terms, a standard deviation *reduction* in union membership (i.e., a −.191 log point change) raises the probability of pro-trade legislative votes by about 25 percentage points. The same *reduction* in union strength decreases the probability of compensation votes by about 7 percentage points.

We again show estimates from standard IV models in columns (1)–(4), which start with an empty model and sequentially enter district controls. The estimates are smaller compared to the dML-IV model. Focusing on the specification (4) with full controls, the magnitude of the estimates relative to their (cluster-robust) standard errors indicates that they remain substantively and statistically significant.[11] As in the previous analysis, the effect of unions remains significant if we allow for a local violation of the exclusion restriction. In additional robustness tests, we account for offshorability, foreign direct investment, district social capital, and the historical size of the industrial base (SI Section D.1, pp. 12–13). We also study results using an alternative estimator (SI Section D.3, pp. 13–14). Our results are also consistent when analyzing votes separately (SI Figure D.1) and adjusting for the share of public unions (SI Table H.1).

[11] The effect of an SD decrease in union strength on trade votes is about 9 percentage points, while the effect on compensation votes is about −5 points.

TABLE 3 Effect of union density on compensation roll-call votes.

	Instrumental variable (IV)				Debiased machine learning IV (dML-IV)[a]
	(1)	(2)	(3)	(4)	(5)
Union density [log]	.266	.205	.205	.246	.376
	(.060)	(.057)	(.080)	(.074)	(.179)
IV violation robust [p][b]	.000	.003	.045	.021	–
Observations	5,458	5,458	5,454	5,380	5,380
District-level controls:					
Slavery and immigration history		✓	✓	✓	✓
Demographics and ideology			✓	✓	✓
Technological change				✓	✓

Notes: Two-stage least squares (2SLS) estimates of IV models with state-congress fixed effects and cluster-robust standard errors (in parentheses). Controls are as in Table 1. Union density is instrumented by share of mining employment in the 1950s. First-stage robust F statistics range from 26.4 to 51.3. Supporting Information (SI) Table I.1 presents full table including inference robust to potentially weak instruments.

[a]Partial linear IV model estimated using debiased machine learning with sample splitting (100 sets) and fivefold cross-fitting (Chernozhukov et al., 2018). Estimates and standard errors calculated using the median method. See SI Section F for details.

[b]Test of union coefficient allowing for local-to-zero violation of IV exclusion restriction, H_0: $\beta = 0$ w. $Cov(Z,e) \approx 0$ (Wang et al., 2018).

Decomposing the effect of trade shocks

Now we turn to the results of the effect decomposition that estimates the full chain from trade exposure to legislative votes working through district-level union membership as well as the direct effect not working through unions. Table 4 shows the resulting estimates from the semiparametric IV model for causal mediation that, for identification, leverages separate IVs for trade exposure and unionization. The table entries are estimates of the LATE of import exposure on legislative votes, the (natural) indirect effect (NIE), operating through union density, and direct effect (NDE), and their corresponding standard errors. Finally, the percent column indicates the share of the total effect of trade shocks (among the treated) that is mediated by union membership. All specifications include historical controls measuring the district-level shares of Blacks and foreign-born in 1910. District-level socio-demographics and political ideology, as well as automation risk are added sequentially.

For roll-call votes related to compensation, all specifications indicate that the estimate of the indirect effect is negative, in line with our two prior analyses. Given the size of the effects relative to their standard errors, we reject the null hypothesis of no indirect union effect. Substantively, the most conservative estimate including all controls (specification 3) suggests that a per-worker increase in Chinese import exposure experienced at the median (around \$1,000) compared to no increase reduces the probability of a pro-compensation vote by about 7 percentage points,

TABLE 4 Import exposure, unions, and legislative votes: Indirect and direct effect based on semiparametric instrumental variable (IV) model with two instruments.

	Compensation votes				Trade votes			
	Local average treatment effect (LATE)	Natural indirect effect (NIE)	Natural direct effect (NDE)	%	LATE	NIE	NDE	%
With history controls	–.110	–.079	–.076	51	.127	.062	.125	33
	(.022)	(.015)	(.034)		(.024)	(.017)	(.040)	
Adding ideology and demographics	–.107	–.074	–.060	55	.126	.060	.107	36
	(.021)	(.020)	(.043)		(.025)	(.021)	(.043)	
Adding automation	–.102	–.067	–.062	52	.123	.052	.103	33
	(.022)	(.019)	(.043)		(.023)	(.019)	(.037)	

Notes: Entries are estimates (and standard errors) of the treatment effect (i.e., LATE) and natural indirect and direct effects among the treated (NIE(1) and NDE(1)). The % column shows how much of the total effect among the treated on votes is mediated via union density (in percent). All specifications include state fixed effects (removed via orthogonal projection). Controls are the same as in Table 1. All models include district-level historical controls for slavery and immigration. The current sets of controls (pre-shock demographics and ideology in row 2, technological change in row 3) are added sequentially. Entries are robust statistics (median and mean absolute deviation) based on 1,999 bootstrap draws. Based on semiparametric IV estimates (Frölich & Huber, 2017); see Supporting Information (SI) Section G.1 for model and estimation details.

on average, *through* its depressing effect on unions. The estimate for the direct effect, which fixes unionization to the level without an increase in import exposure, is slightly smaller (and much less precisely estimated). About half—between 51% and 55%—of the total effect of district-level trade exposure on compensation votes is mediated by union membership. The estimates imply that trade-induced weakening of unions as a countervailing power is a large factor explaining the lack of political support for increased compensation in the face of a historic shift in the pattern of global trade generating many losers in the American economy.

For votes on trade policy, the indirect effect working through unions is positive in all specifications, in line with our prior analysis. But we are now able to quantify the relative magnitude of the indirect and direct effects. The most conservative estimate of the NIE is from the specification using all controls. It implies that a $1,000 per-worker increase in Chinese import exposure increases the probability of a free-trade vote by 5 percentage points by weakening local unions. The estimates of the NDE not working through unions is also positive (in contrast to the NDE of compensation votes, it is more precisely estimated). The union channel accounts for about a third (33%–36%) of the effect of import exposure on how legislators vote on trade policy. This means that import exposure can increase support for further trade liberalization by undermining local unions, one of the few domestic countervailing powers increasing the political clout of non-elite workers.[12]

Additional analyses in SI Table G.1 further probe the robustness of the results. We adjust for offshorability, social capital, for historical industrial employment, for the contemporary industry mix, and for a proxy of the pre-trade shock distribution of union membership. We use Rotemberg weights to identify and exclude influential industries. We also show that estimates are not driven by influential roll-calls (SI Figure G.1).

CONCLUSION

While many scholars in the recent literature have studied the effects of import exposure on democratic politics, they have paid less attention to labor unions as a mechanism. We take up the quest for mechanisms using a double IV approach for causal mediation that enables us to disentangle and quantify the mediating effect of unions. Our analysis revealed that labor unions are a central mechanism linking the steep rise in international economic competition from China after 1989 and the domestic politics of trade and compensation in the American political economy.

While earlier scholarship on globalization and labor unions correctly pointed out that there is nothing inevitable about the impact of globalization on unions, we demonstrate that regional trade shocks did substantively reduce unionization at the local level in the United States, with important consequences for the political representation of economic losers. By weakening unions as a countervailing power shaping the political representation of non-elite workers in Congress, import competition reduced politicians support for economic policies helping those most affected by the distributive effects of trade. Through weaker unions, import competition also increased legislative support for additional trade liberalization, contributing to the hyperglobalization apparent before the onset of the trade wars during Donald Trump's presidency. Overall, global competition has undermined political support for embedded liberalism in the twenty-first century to a significant degree by weakening unions.

The institutional channel highlighted in our findings suggests that the political impact of international trade is likely to last beyond the electoral cycles when its economic impact was felt most acutely. Given the costs and uncertain prospects of trying to unionize the workplace, a large negative shock to union membership is difficult to offset, at least in the short run. The weakening of unions may also tilt partisan balance in domestic politics, thus altering representation across a much wider range of issues.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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